



ECE 5300: Electronic Circuits II

Heart Rate Monitor Project

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March 31, 2026

Table of Contents

1.0	Executive Summary	1
2.0	Introduction.....	1
2.1	Background on Heart Rate Monitoring.....	1
2.2	Project Objectives	2
2.3	Overview of System Design	2
3.0	System Design and Methodology	2
3.1	Overall Design Approach.....	2
3.2	System Block Diagram	3
3.3	Signal Flow Description	4
4.0	Sensor and Raw Signal Analysis.....	5
4.1	Photo-Transistor Operation.....	5
4.2	Raw Signal Characteristics	5
5.0	Sensitivity Analysis.....	6
5.1	Sensitivity to Ambient Light.....	6
5.2	Sensitivity to Sensor Positioning	7
6.0	Signal Conditioning	8
6.1	AC Coupling and High-Pass Filter Design.....	8
6.2	Amplifier Design and Gain Stages	9
6.3	Comparator and Square Wave Generation.....	9
6.4	Conditioned Signal Analysis (Pre vs Post Processing).....	9
7.0	Signal-to-Noise Ratio Analysis.....	10
7.1	Noise Sources.....	10
7.2	Effect of LED Intensity on SNR.....	11
7.3	Noise Mitigation Techniques	11
8.0	Frequency Extraction and Conversion.....	12
8.1	Heart Rate Signal Characteristics	12
8.2	Frequency Detection Method.....	12
8.3	Frequency-to-Voltage Conversion Technique.....	12
9.0	Alarm and Threshold Detection.....	14
9.1	Comparator Design	14
9.2	Upper and Lower Thresholds.....	14

9.3	Alarm Triggering Mechanism.....	15
10.0	Calibration and Error Analysis.....	15
10.1	Calibration Procedure and Measurement Accuracy.....	15
10.2	Sources of Error	16
10.3	Temperature Effects	16
11.0	Long-Term Testing.....	16
11.1	Test Methodology (6-Hour Evaluation).....	16
11.2	Observations Over Time	16
11.3	Stability and Drift Analysis.....	17
12.0	Safety Considerations	17
12.1	Electrical Safety	17
12.2	Optical Safety (LED Exposure)	17
12.3	Isolation and User Protection.....	17
13.0	Practical Considerations.....	17
13.1	Compactness of Design.....	17
13.2	Usability and Ergonomics	19
13.3	Potential Improvements	19
14.0	Development Challenges	20
14.1	Issues Encountered.....	20
14.2	Troubleshooting Process	20
14.3	Lessons Learned.....	21
15.0	Experimental Results and Validation	21
15.1	Output Waveforms	21
15.2	Response to Physical Activity.....	25
16.0	Teamwork and Contributions.....	25
16.1	Task Distribution and Individual Contributions	25
17.0	Conclusion	26
18.0	References.....	27

List of Tables

Table 10.1: Calibration and Measurement Accuracy	16
Table 11.1: 6-hour test results	16
Table 14.1: Major issues encountered along with their severity and impact	20

List of Figures

Figure 3.1: Functional Block Diagram	3
Figure 4.1: Transmissive optical sensing circuit used to generate the raw heart rate signal	5
Figure 4.2: Raw heartbeat signal measured at the collector	6
Figure 5.1: Measuring the output after the initial gain and filtering stage in the enclosure	6
Figure 5.2: Signal after the gain stage, outside enclosure, ambient light conditions.....	7
Figure 5.3: Waveform with fingernail facing the receiver.	7
Figure 5.4: Waveform with fingernail to emitter.	7
Figure 5.5: Waveform with side of finger touching emitter and receiver.	8
Figure 6.1: AC-coupled high-pass filter removing DC offset.....	8
Figure 6.2: Multi-stage heartbeat signal conditioning circuit.....	9
Figure 6.3: Raw signal (pre-processing).....	10
Figure 6.4: Amplified and filtered signal.....	10
Figure 8.1: Signal characteristics.....	12
Figure 8.2: RC circuit for average voltage.....	13
Figure 8.3: Average voltage waveform	13
Figure 9.1: Voltage dividers for threshold voltages	14
Figure 9.2: Dual Comparator Circuit for High and Low Heart Rate Detection	15
Figure 13.1: Circuit and battery used for power	18
Figure 13.2: Enclosure made in CAD.....	18
Figure 13.3: Assembled enclosure	19
Figure 15.1: Circuit stage 1.....	21
Figure 15.2: Output Waveform at Test Point 1	21
Figure 15.3: Stage 2 of Circuit.....	22
Figure 15.4: Output Waveform at Test Point 2	22
Figure 15.5: Stage 3 of Circuit.....	23
Figure 15.6: Output Waveform at Test Point 3	23
Figure 15.7: Stage 4 of Circuit.....	24
Figure 15.8: Output Waveform at Test Point 4	24
Figure 15.9: System output after physical activity	25

1.0 Executive Summary

This report presents the design and implementation of an analog optical heart rate monitor developed for ECE 5300 Electronic Circuits II. The system uses infrared light and a phototransistor to detect changes in blood volume within the user's fingertip, producing a small, noise-sensitive signal.

To extract a reliable heartbeat signal, a multi-stage analog conditioning chain was designed and built. A high-pass filter removes DC offsets that vary between users, while low-pass filters reduce high-frequency noise. The signal is then amplified and centered at 2.5 V to allow proper operation on a single 0–5 V supply.

The conditioned signal is converted into a digital waveform using a comparator, then passed through a 555 timer in monostable mode to generate uniform pulses. A frequency-to-voltage stage produces an average voltage proportional to heart rate, which is compared against thresholds for low and high BPM. LEDs provide real-time feedback for heartbeat detection and heart rate range.

System performance was validated against an Apple Smart Watch, with a maximum observed error of 3 BPM across various conditions. Long-term testing confirmed stable operation, and the system remained reliable under different lighting conditions and finger placements.

The final design is packaged in a compact, battery-powered enclosure. Development challenges included buzzer loading, sensor sensitivity, and RC tuning, which were resolved through iterative testing.

Overall, the project met all objectives and demonstrates the effectiveness of modular analog signal processing for heart rate monitoring. Future improvements include transitioning to a custom PCB, further noise reduction, and refining the enclosure design.

2.0 Introduction

2.1 Background on Heart Rate Monitoring

Optical heart rate monitoring is commonly implemented using photoplethysmography, which is a technique that detects variations in blood volume within tissue using light. In this method, an LED illuminates the skin while a phototransistor detects transmitted or reflected light. As the heart pumps blood, the blood volume in the fingertip changes, altering the absorbed light intensity and producing a pulsatile electrical signal corresponding to the cardiac cycle [1], [2]. This signal contains a small AC component representing a heartbeat and a larger DC component caused by tissue absorption and ambient light.

Due to the AC component being significantly smaller than its DC counterpart, signal conditioning is required to extract meaningful information. High-pass filtering removes the DC component, while amplification and additional filtering improve signal-to-noise ratio and reduce

interference from ambient light and motion [1], [3]. These techniques are widely used in optical heart rate monitoring and biomedical instrumentation systems.

2.2 Project Objectives

The objective of this project is to design and implement an optical heart rate monitor using analog electronic circuits. The system detects heartbeats from a fingertip sensor, conditions the signal, and extracts heart rate information for monitoring and alarm detection.

The project goals include:

- I. Design of an optical LED and phototransistor sensor
- II. Amplification and filtering of the raw signal
- III. Extraction of heartbeat frequency
- IV. Frequency-to-voltage conversion
- V. Implementation of high and low heart rate alarms
- VI. Evaluation of system accuracy, stability, and safety

The system is also tested under varying ambient light conditions, sensor positioning, and long-term operation to ensure reliable performance.

2.3 Overview of System Design

The heart rate monitor consists of multiple analog signal-conditioning stages designed to extract the pulsatile component of the optical sensor signal. An LED and phototransistor placed on the fingertip detect variations in blood volume, producing a small AC signal corresponding to each heartbeat.

The raw sensor output is first AC-coupled to remove the DC component and slow variations caused by ambient light and tissue absorption. The signal is then passed through multiple amplification and filtering stages to increase signal amplitude and reduce noise. These operational amplifier stages condition the signal, producing a clearer waveform corresponding to each heartbeat.

After conditioning, the processed waveform is suitable for comparator-based pulse detection, allowing the heartbeat frequency to be measured and converted into heart rate. This modular design allows each stage to be tested independently and improves overall system reliability.

3.0 System Design and Methodology

3.1 Overall Design Approach

The system is designed using a modular analog signal processing approach to detect heart rate. An infrared (IR) LED and phototransistor detect changes in blood flow within the user's finger, producing a small analog signal.

To ensure reliable operation, this signal is conditioned through multiple stages of amplification, filtering, and buffering to improve its quality and stability. These stages work together to remove unwanted noise, eliminate DC offsets, and increase the signal amplitude to a level suitable for detection. A 2.5 V reference is used to center the signal within the 0–5 V supply range, enabling proper single-supply op-amp operation while maintaining room for signal variations and improving consistency across different users.

The conditioned signal is then converted into a digital pulse using a comparator, which generates a transition each time a heartbeat is detected. This output is passed to a 555-timer configured as a pulse-shaping circuit, ensuring that each detected heartbeat produces a clean, uniform pulse regardless of variations in the input waveform.

Finally, the system provides both real-time and averaged feedback through LEDs. A direct output indicates each heartbeat as it occurs, while an additional averaging stage produces a signal proportional to heart rate, enabling high and low BPM detection through threshold-based indicators.

3.2 System Block Diagram

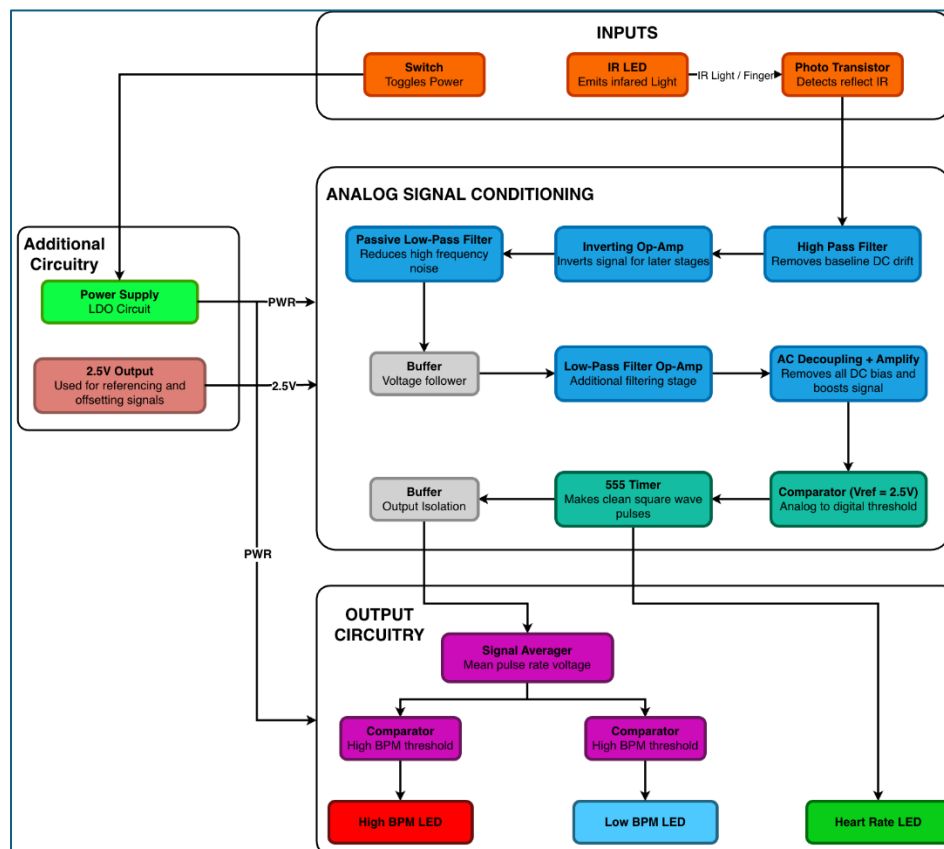


Figure 3.1: Functional Block Diagram

3.3 Signal Flow Description

The system begins with an infrared (IR) LED emitter, which transmits light into the user's finger. A phototransistor detects the reflected infrared light, producing a small analog signal that varies with changes in blood flow associated with each heartbeat.

This raw signal is passed through a series of signal conditioning stages to improve its quality and prepare it for digital processing. First, a high-pass filter removes DC offsets caused by differences between users, as each individual may introduce a different baseline signal level, as well as variations in finger placement and lighting conditions. The signal is then inverted using an operational amplifier allowing later stages to restore the polarity while also being biased to approximately 2.5 V to center it within the 0 – 5 V supply range. This signal then flows into a passive low-pass filter, which reduces high-frequency noise.

The signal is then routed through a buffer (voltage follower) to provide impedance isolation between stages. Next, an active low-pass filter (op-amp based) further refines the signal by reducing remaining noise components. An AC coupling and amplification stage is then used to remove all DC bias (including our 2.5V bias) and increase the signal amplitude, ensuring reliable threshold detection.

The conditioned analog signal is then applied to a comparator, where it is compared against a 2.5V reference voltage. This converts the analog waveform into a digital pulse corresponding to each detected heartbeat. The comparator output is then fed into a 555-timer configured as a pulse-shaping circuit, which produces clean and consistent square wave pulses. A final output buffer ensures that the signal can drive subsequent stages without loading the timer or other circuitry.

At the output stage, the signal is divided into multiple paths. One path drives a heart rate indicator LED directly from the output of the 555, which flashes in real time with each detected heartbeat. A second path feeds a signal from the buffer into a voltage averaging circuit, which generates a voltage proportional to the average pulse rate. This averaged signal is then compared against predefined thresholds using two comparators, one for high BPM detection and one for low BPM detection. Each comparator drives a corresponding LED indicator, providing visual feedback when the heart rate exceeds or falls below the desired range.

Additionally, the system is powered through a switch-controlled supply, which feeds into a low-dropout (LDO) regulator. The LDO provides a stable supply voltage to all analog and output stages, ensuring consistent operation. From this regulated supply, the 2.5 V reference voltage used by the comparator is also derived, enabling accurate and stable threshold detection throughout the system.

4.0 Sensor and Raw Signal Analysis

4.1 Photo-Transistor Operation

The optical sensing circuit used to generate the raw heartbeat signal is shown in Figure 4.1. An infrared LED (LTE-302) is used as the emitter, while a phototransistor (LTR-301) acts as the receiver. The LED is driven by a 5 V supply through a current limiting resistor that is $470\ \Omega$, ensuring a consistent infrared illumination through the fingertip. The phototransistor is configured in a common-emitter configuration with a $1\ \text{k}\Omega$ pull up resistor.

When a finger is placed between the emitter and the receiver, variations in blood volume in the tissue during the cardiac cycles alter the amount of transmitted light. These variations produce a small voltage signal at the collector of the phototransistor. This signal is the raw heartbeat signal and will be manipulated in further stages of the system.

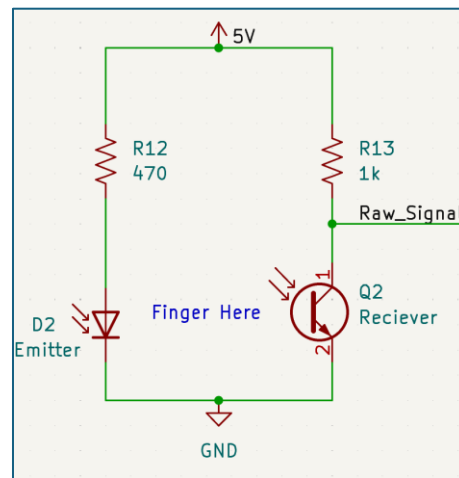


Figure 4.1: Transmissive optical sensing circuit used to generate the raw heart rate signal

4.2 Raw Signal Characteristics

The raw signal shown in Figure 4.2 shows a small AC component superimposed on a noisy baseline. The heartbeat signal amplitude ranges approximately from $+10\ \text{mV}$ to $-5\ \text{mV}$, making it difficult to clearly distinguish individual heartbeats. Significant noise is also present due to ambient light, sensor placement, and electronic interference. Due to this, further filtering and amplification stages were necessary to improve the signal-to-noise ratio and make the heartbeat waveform more distinguishable for subsequent processing.

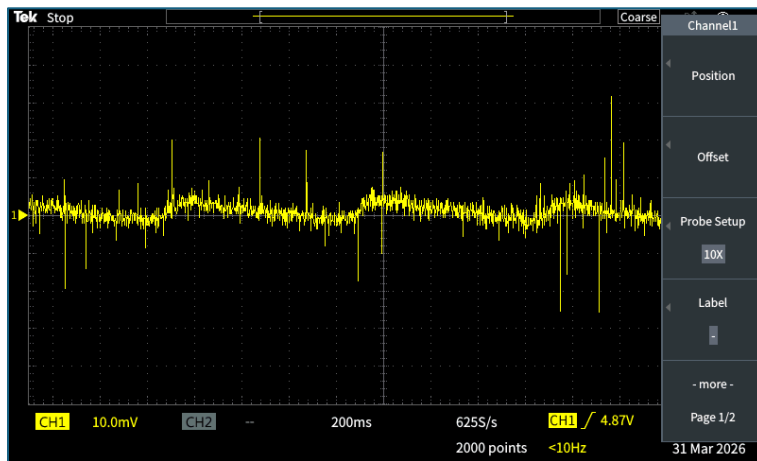


Figure 4.2: Raw heartbeat signal measured at the collector

5.0 Sensitivity Analysis

5.1 Sensitivity to Ambient Light

To measure sensitivity to lighting conditions, 2 heartbeat measurements were taken under external lighting and within the enclosure to compare any differences. From Figures 5.1 and 5.2, it can be seen that in ambient lighting conditions, the noise floor was slightly higher, with additional peaks observed separate from the heartbeat. However, with proper filtering in the gain stage, the heart rate pulses remain the strongest signal in both conditions.

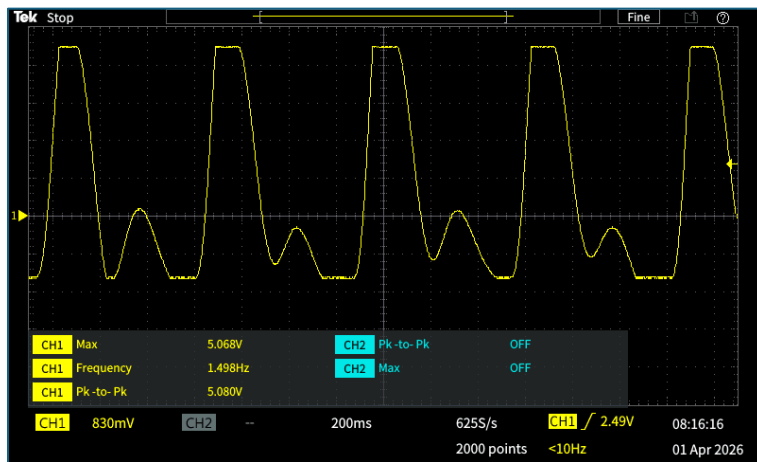


Figure 5.1: Measuring the output after the initial gain and filtering stage in the enclosure

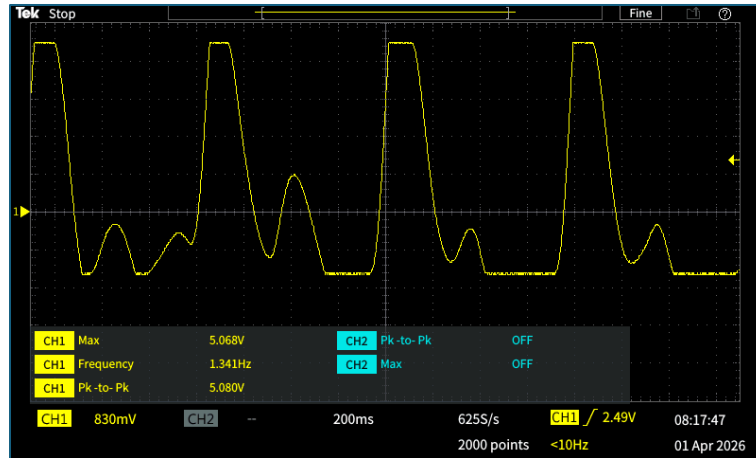


Figure 5.2: Signal after the gain stage, outside enclosure, ambient light conditions

5.2 Sensitivity to Sensor Positioning

To determine variation in finger position within the enclosure, three samples were taken with the finger in different positions, as shown in Figures 5.3-5.5. When the nail is facing either the emitter or the receiver, the signal is strong enough to be amplified to the op-amp rail, as shown by the clipping in 5.3 and 5.4. This signal is slightly weaker when the sides of the finger are placed at both the emitter and the receiver, as shown in Figure 5.5, where the heartbeat is strong but not driven to the rail.

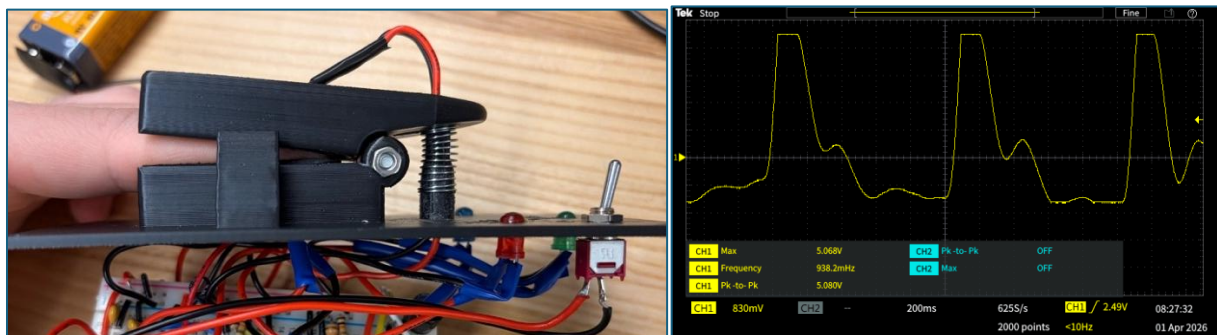


Figure 5.3: Waveform with fingernail facing the receiver.

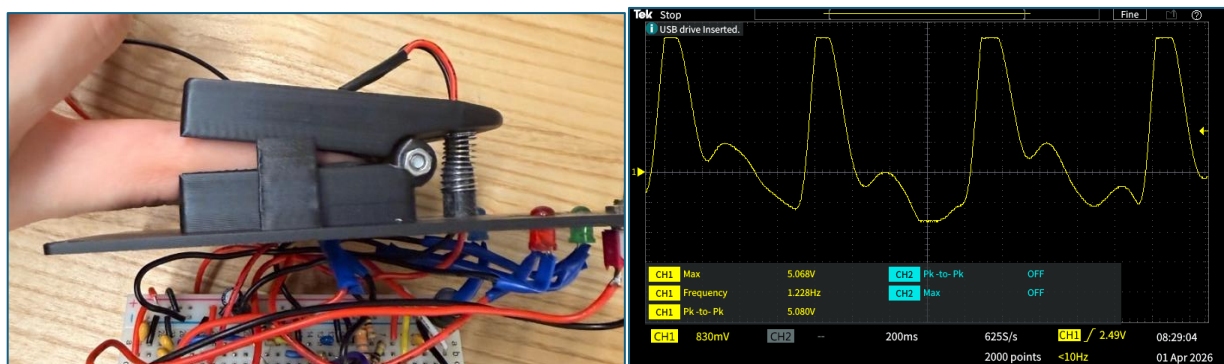


Figure 5.4: Waveform with fingernail to emitter.

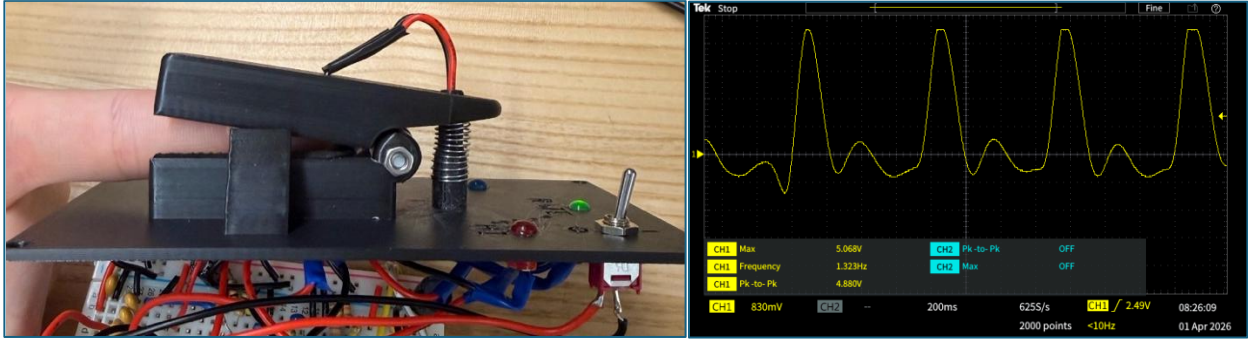


Figure 5.5: Waveform with side of finger touching emitter and receiver.

6.0 Signal Conditioning

6.1 AC Coupling and High-Pass Filter Design

The raw sensor output contains a small AC component corresponding to the heartbeat and a large DC offset and low-frequency variations. Before using the signal for anything useful, this DC offset must be removed; this was done using a passive high-pass filter. To ensure the actual heartbeat remained unchanged, the filter was designed to block only frequencies below 3.4 mHz.

A 2.5V offset was also added at this stage, this was required because the MCP6002 operational amplifiers used were all powered using a single supply from 0 V to 5 V. Biasing the signal at mid-supply allowed both positive and negative variations of the heartbeat signal to be amplified in later stages without clipping at the rails.

The high-pass filtering stage also reduces slow variations caused by ambient light and motion artifacts. The filtered output is then passed to subsequent amplification stages for further conditioning.

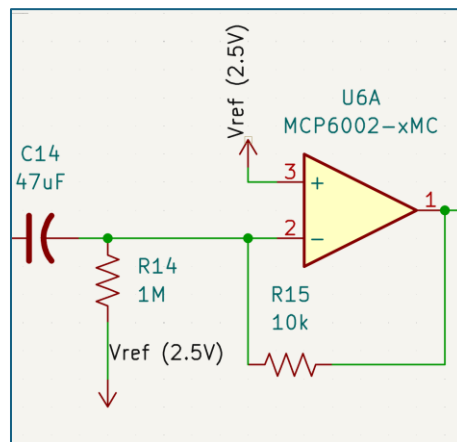


Figure 6.1: AC-coupled high-pass filter removing DC offset

6.2 Amplifier Design and Gain Stages

After the initial AC-coupled high-pass filter, the signal was passed through a passive low-pass filter (3.4 Hz) to reduce high-frequency noise. A buffer stage was then used to isolate the filter and prevent loading effects. The signal was then amplified using an active gain stage (gain of 470) with additional filtering in the feedback network (3.4 Hz) to further improve waveform clarity. Following this, the signal was AC-coupled again to remove any accumulated DC offset.

Finally, a second amplification stage (gain of 11) increased the signal amplitude before the conditioned waveform was sent to the comparator and 555 timer for pulse generation.

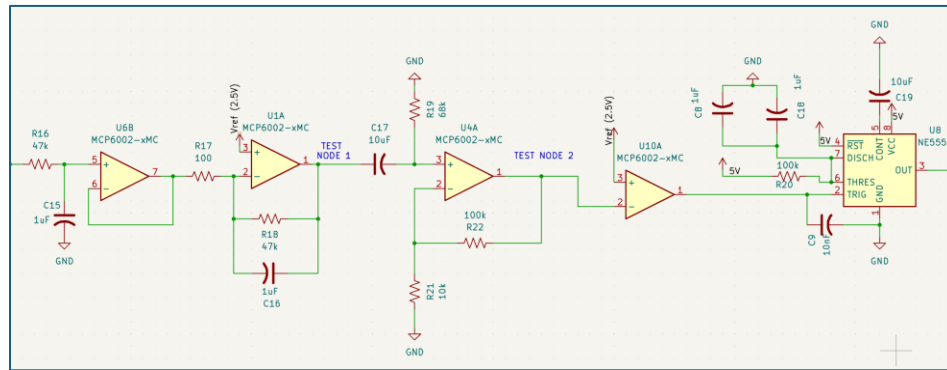


Figure 6.2: Multi-stage heartbeat signal conditioning circuit

6.3 Comparator and Square Wave Generation

Following the amplification stages described in Section 6.2, the conditioned heartbeat signal was converted into a square wave using a comparator. The reference voltage was set to 2.5 V, matching the mid-supply bias used throughout the signal conditioning stages. This allowed the circuit to detect when the amplified waveform crossed the midpoint and generate a pulse for each heartbeat.

The comparator output was then fed into a 555-timer configured to produce a fixed pulse width of approximately 248 ms. This provided consistent pulse generation and prevented multiple triggers from a single heartbeat.

6.4 Conditioned Signal Analysis (Pre vs Post Processing)

The raw heartbeat signal was initially difficult to interpret due to low amplitude and noise. After signal conditioning, the waveform became a clear square wave representing the heartbeat frequency.

Figure 6.3 shows the raw signal, while Figure 6.4 shows the conditioned square wave output. This improved waveform allowed the heartbeat frequency to be clearly identified and used for accurate BPM calculation.

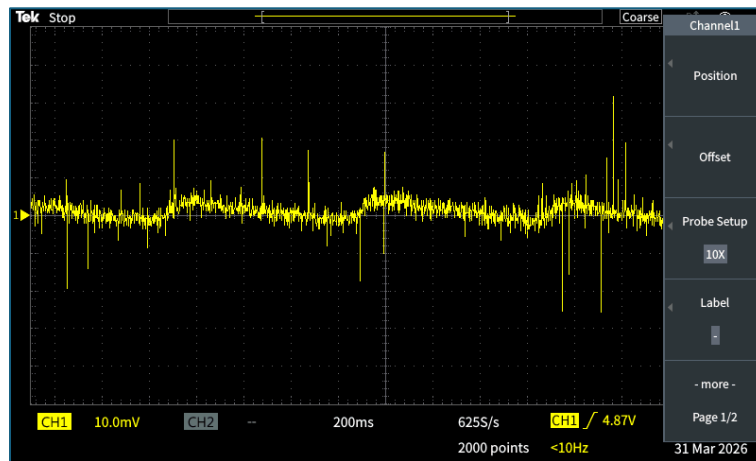


Figure 6.3: Raw signal (pre-processing)

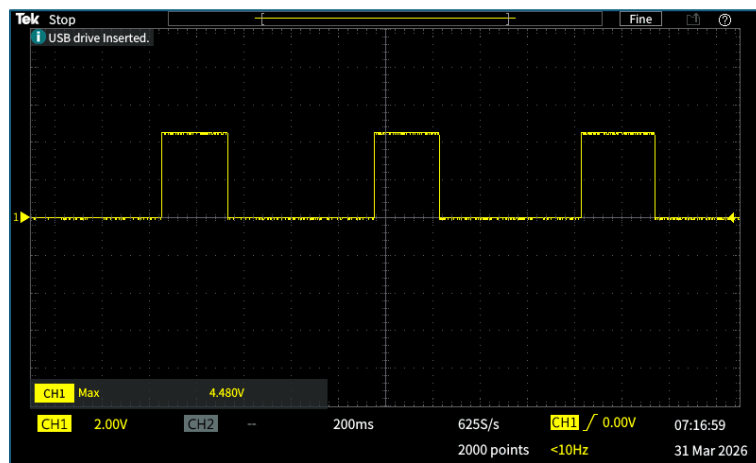


Figure 6.4: Amplified and filtered signal

7.0 Signal-to-Noise Ratio Analysis

7.1 Noise Sources

The raw heartbeat signal shown has a low-amplitude waveform with noticeable noise and spikes. The desired signal (periodic variation due to blood flow) is relatively small compared to the overall noise level, making signal conditioning necessary to reliably extract heart rate information.

Several noise sources contribute to the signal-to-noise ratio (SNR):

- **Ambient Light Interference:** External light sources can introduce fluctuations in the phototransistor output, especially if the finger enclosure is not fully sealed. Changes in room lighting or movement can increase this effect.

- **Motion:** Small movements of the finger during measurement cause variations in light absorption, leading to low-frequency noise and spikes in the signal.
- **Power Supply Noise:** The use of a 9V battery and voltage regulation stages can introduce noise, particularly if decoupling is insufficient. Any ripple or instability in the supply can propagate through the analog stages.
- **Photodetector Noise:** The phototransistor itself introduces shot noise and sensitivity variations depending on light intensity.
- **Electrical Interference:** Nearby electronic devices and wiring can introduce electromagnetic interference (EMI), which appears as high-frequency noise.

7.2 Effect of LED Intensity on SNR

The LED intensity directly affects the system's SNR, as it determines the strength of the optical signal received by the phototransistor. As the LED intensity increases, more light comes through the finger and is modulated by blood flow, resulting in a larger AC signal component corresponding to the heartbeat.

At low LED intensities, the detected signal is very small and becomes comparable to noise sources such as ambient light, electronic noise, and motion. This results in a poor SNR, making it difficult to clearly distinguish heartbeat peaks from noise.

As the LED intensity is increased to an appropriate level, the amplitude of the desired signal increases while the noise remains relatively constant. This improves the SNR and makes the heartbeat waveform more distinguishable, allowing for more reliable detection and processing.

However, if the LED intensity is too high, the phototransistor may begin to saturate, reducing sensitivity to small changes in light caused by blood flow. Additionally, higher intensities can increase power consumption and may introduce additional noise or heating effects. Therefore, an optimal LED intensity must be selected to maximize signal strength without causing saturation or excessive noise.

7.3 Noise Mitigation Techniques

To reduce noise, several filtering stages were implemented, including both high-pass and low-pass filters. The high-pass filter removes DC offsets and lower frequencies caused by constant light absorption and user variation, while the low-pass filter reduces high-frequency noise. Signal amplification is distributed across stages, and the signal is biased to 2.5 V for proper single-supply operation. A comparator is then used to convert the conditioned signal into a clean digital pulse, improving robustness to remaining noise.

Further details on the filter design and their role within each stage of the system can be found in Section 3.

8.0 Frequency Extraction and Conversion

8.1 Heart Rate Signal Characteristics

The final heartbeat signal consists of a square wave with a peak voltage of 4.48 V and a pulse width of approximately 248 ms. Each pulse corresponds to a detected heartbeat, producing a periodic waveform suitable for frequency measurement. As shown in Figure 8.1, the signal maintains a consistent amplitude and pulse width, providing a stable waveform for accurate heart rate extraction and further processing.

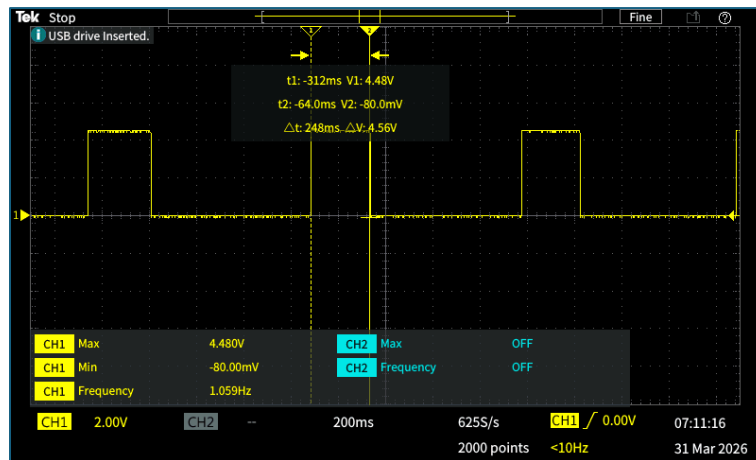


Figure 8.1: Signal characteristics

8.2 Frequency Detection Method

The frequency of the heartbeat signal was determined by measuring the period between successive square wave pulses. Since each pulse corresponds to one detected heartbeat, the heart rate frequency can be extracted directly from the timing of the waveform.

The square wave produced in the previous stage simplifies frequency detection by providing clear rising edges that represent individual heartbeat events. By measuring the time between these pulses, the signal period can be determined and used for heart rate calculation in the following section.

In the final unit, an LED is also connected to the square wave output so that it flashes once per detected heartbeat. This provides a visual indication of heart rate operation and confirms proper heartbeat detection during testing and use.

8.3 Frequency-to-Voltage Conversion Technique

An RC circuit was used to convert the square wave frequency into an average voltage. This method works by allowing the capacitor to charge during the high portion of the pulse and discharge during the low portion. As the frequency of the input signal changes, the amount of charge stored on the capacitor also changes, resulting in a different average voltage level.

The average voltage can be approximated using the relationship between peak voltage, pulse width, and frequency. This relationship is shown below:

$$V_{avg} = V_{peak} \cdot t_w \cdot f$$

The RC network used for this conversion is shown in Figure 8.2, where the 555-timer output is passed through a 220 kΩ resistor and 33 μF capacitor. The voltage measured at this node represents the average value of the square wave and is directly related to the heartbeat frequency.

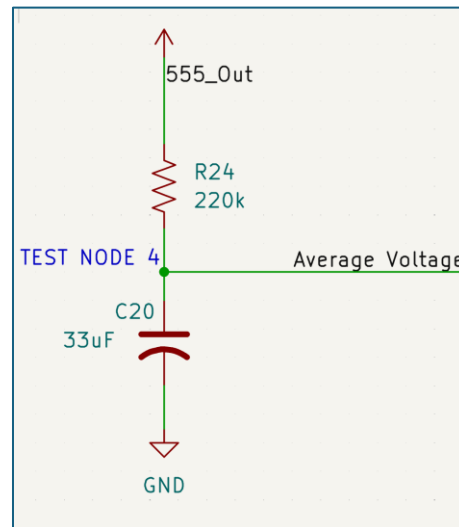


Figure 8.2: RC circuit for average voltage

The resulting average voltage waveform observed on the oscilloscope is shown in Figure 8.3. As the heartbeat frequency increases, the capacitor has less time to discharge, causing the average voltage to rise. Conversely, lower heart rates result in a lower average voltage. This voltage-based representation of frequency was then used for further processing and threshold detection.

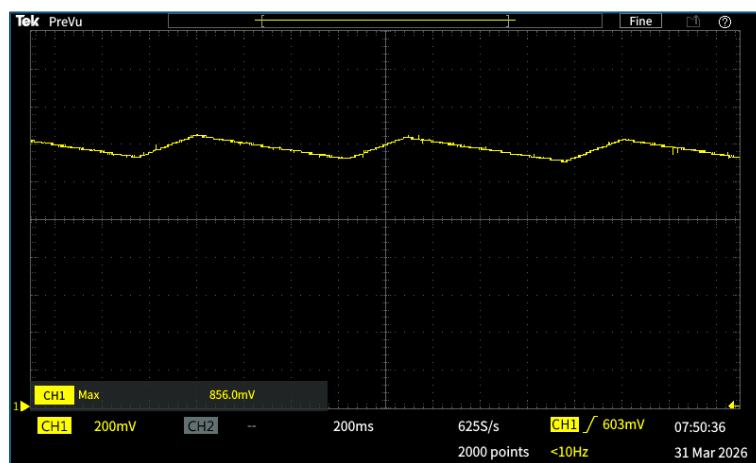


Figure 8.3: Average voltage waveform

9.0 Alarm and Threshold Detection

9.1 Comparator Design

The average voltage produced in the previous stage was used as the input to comparator circuits to determine whether the heart rate falls within an acceptable range. Using the relationship presented in Section 8.3, the average voltages corresponding to 50 BPM and 125 BPM were calculated. These values define the lower and upper limits of the acceptable heart rate range.

The calculations for the upper and lower thresholds are shown below:

$$f_{50} = \frac{50}{60} = 0.833 \text{ Hz}$$

$$V_{avg_{50}} = V_{peak} \cdot t_w \cdot f = 4.48 \cdot 0.248 \cdot 0.833 = \boxed{0.925 \text{ V}}$$

$$f_{125} = \frac{125}{60} = 2.083 \text{ Hz}$$

$$V_{avg_{125}} = V_{peak} \cdot t_w \cdot f = 4.48 \cdot 0.248 \cdot 2.083 = \boxed{2.314 \text{ V}}$$

Once the threshold voltages were determined, the average voltage signal was compared against these reference levels using two comparator circuits. One comparator detects when the heart rate exceeds the upper threshold, while the other detects when it falls below the lower threshold.

9.2 Upper and Lower Thresholds

The calculated threshold voltages were implemented using voltage divider networks, as shown in Figure 9.1. These dividers generate stable reference voltages used by the comparator circuits to determine whether the heart rate falls outside the desired operating range.

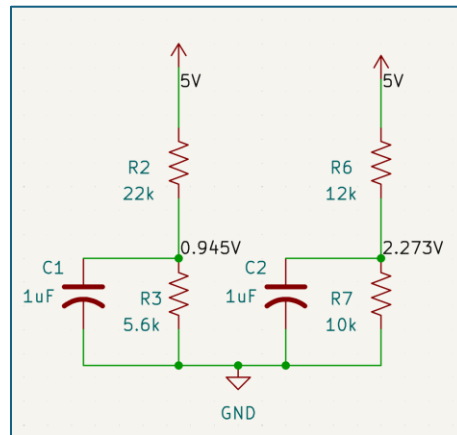


Figure 9.1: Voltage dividers for threshold voltages

Based on the calculations in Section 9.1, the lower threshold corresponds to approximately 0.925 V for 50 BPM, while the upper threshold corresponds to approximately 2.314 V for 125 BPM. Due to the availability of standard resistor values, the exact calculated voltages were not used.

Instead, the voltage divider networks were designed using available resistors to produce approximate threshold voltages of 0.925 V and 2.314 V.

Capacitors were included at each reference node to reduce noise and stabilize the threshold voltages. These filtered reference voltages were then applied to the comparator inputs, allowing detection when the average voltage rises above the upper threshold or falls below the lower threshold.

9.3 Alarm Triggering Mechanism

The outputs from the comparators were used to drive LED indicators for abnormal heart rate detection. As shown in Figure 9.2, one comparator output activates the high heart rate indicator, while the second comparator output activates the low heart rate indicator.

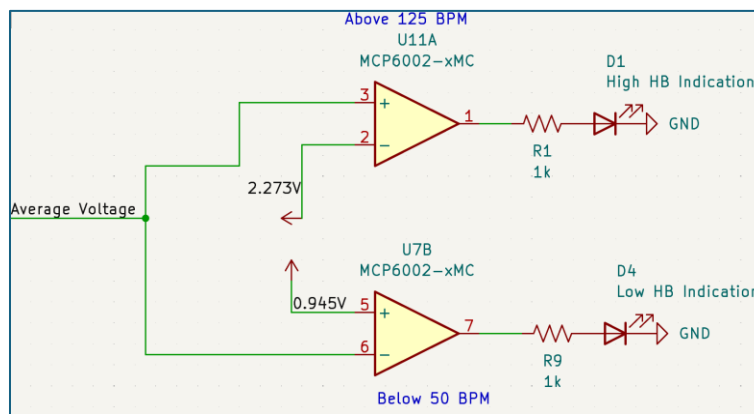


Figure 9.2: Dual Comparator Circuit for High and Low Heart Rate Detection

The diode configuration of the comparator outputs functions as an OR gate, allowing both abnormal conditions to activate the alarm indication. A red LED indicates a heart rate above the upper threshold, while a blue LED indicates a heart rate below the lower threshold. Due to reliability issues encountered with the buzzer, LED indicators were selected as a more stable and practical method for alarm indication.

10.0 Calibration and Error Analysis

10.1 Calibration Procedure and Measurement Accuracy

The heart rate monitor was calibrated by comparing measured values with a reference device (Apple Smart Watch) under various conditions. Measurements were taken at rest, during mild activity, and after exercise to evaluate system performance across a range of heart rates. The LED heartbeat indicator and square wave output were also monitored to ensure each detected pulse corresponded to a heartbeat.

Table 10.1: Calibration and Measurement Accuracy

Condition	Reference BPM	Measured BPM	Average Voltage (V)	Error (BPM)
Resting	82	84	1.56	2
Mild Activity	95	98	1.82	3
After Exercise	145	147	2.72	2
Recovery	108	111	2.06	3

The measured values were compared to the reference device, and the results are summarized in Table 10.1. The system produced consistent readings across different operating conditions, with only minor deviations. These variations were primarily due to sensor positioning and minor signal fluctuations.

10.2 Sources of Error

Some sources of error come from finger placement, small body movements, and noise in the signal itself caused by external light. The amount of light reaching the transceiver varied with the finger's position, affecting the reading. Movement during testing could also make the signal less steady and slightly change the BPM. Some errors may also have come from component tolerances, which would affect the reference voltages provided by VDRs. Even with all these sources of error, our results remained very accurate.

10.3 Temperature Effects

Temperature can affect the system slightly by changing how the LED and photo transistor behave. It can also slightly alter the signal level, which may result in small changes in the measured heart rate. During normal testing, we didn't notice any change over time, but this may be due to the small sample size. Long term testing was conducted in section 11 to confirm accuracy.

11.0 Long-Term Testing

11.1 Test Methodology (6-Hour Evaluation)

We left our original circuit on the breadboard, with 5 V supplied by the power supply, for approximately 6 hours. We measured our resting heart rate at the top of each hour to ensure accuracy over time. The square-wave output and the general system response were recorded.

11.2 Observations Over Time

Table 11.1: 6-hour test results

Time (hr)	Reference BPM	Measured BPM	Average Voltage (V)	Error (BPM)
0	82	84	1.56	2
1	80	83	1.54	3
2	78	82	1.52	4

3	78	81	1.50	3
4	78	82	1.52	4
5	77	81	1.50	4
6	77	83	1.54	6

11.3 Stability and Drift Analysis

Overall, the system showed reasonable stability during long-term testing. A small discrepancy was noticed over time, which may be due to heating, small changes in sensor position, or changes in ambient light and surroundings. However, these discrepancies weren't significant enough to affect the system's functionality, suggesting the design is stable enough for basic continuous use, though further improvement could be made.

12.0 Safety Considerations

12.1 Electrical Safety

Since this device is used directly on the fingertip, electrical safety was taken into account to ensure no harm to the user. The circuit operates from a low-voltage 5 V LDO powered by a 9 V battery. Current-limiting resistors were also used in series with the LEDs and IR RX/TX to prevent excess current flow and overheating. While we were testing the circuit, we verified each stage was operating correctly by implementing standard test nodes to confirm the system was functioning as expected.

12.2 Optical Safety (LED Exposure)

The receiver uses IR light passing through the fingertip, so optical safety was also considered during operation. Although the light was only operated at a low voltage, direct eye exposure was avoided during testing. The emitter is out of sight in our final design to avoid direct eye exposure.

12.3 Isolation and User Protection

User protection was improved in our final design by adding a neoprene foam spring to the clamp, allowing for secure finger placement. This allows for a secure design without applying too much pressure to the user's finger. We also considered using a nonconductive enclosure, which ensures minimal emitted light leaves the enclosure while isolating the user from the circuit itself.

13.0 Practical Considerations

13.1 Compactness of Design

The system was designed with an emphasis on compactness to improve portability and overall integration. However, due to time constraints on the project, implementing a custom printed

circuit board (PCB) was not feasible. Instead, the circuit was built on a half-sized breadboard folded to vertically stack its sections, significantly reducing the footprint while maintaining full functionality, as shown in Figure 13.1. Additionally, the system is powered by a 9 V battery, enabling its portability.

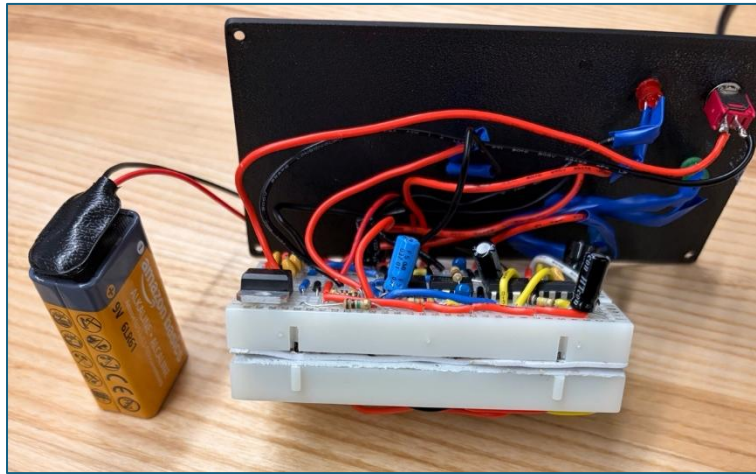


Figure 13.1: Circuit and battery used for power

A custom enclosure was developed to house the folded breadboard and its battery. The enclosure minimizes unused internal volume and securely holds the breadboard, resulting in a compact, organized, and handheld heart rate monitor design. Figure 13.2 shows the isometric view of the CAD assembly.

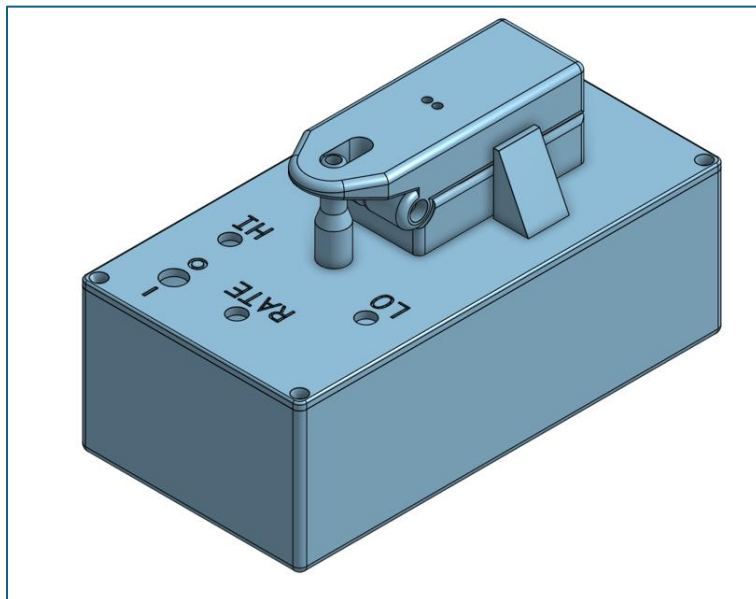


Figure 13.2: Enclosure made in CAD

13.2 Usability and Ergonomics

The system is designed to be simple and intuitive to use. The enclosure includes cutouts and labeling for the power switch and three LEDs corresponding to heart rate, low BPM, and high BPM. This allows for easy control and clear visual feedback for the user.

The finger placement mechanism is integrated into the top lid, securing the phototransistor and photodetector for consistent alignment. A spring holds the finger clamp in place, ensuring stable contact and improving measurement reliability across different users. The final construction is shown in Figure 13.3 below.



Figure 13.3: Assembled enclosure

13.3 Potential Improvements

A key improvement to the current physical design and layout would be transitioning from a breadboard implementation to a custom printed circuit board. This would significantly reduce the overall size of the system, as components could be more tightly integrated without the spacing and wiring constraints of a breadboard.

From an enclosure perspective, a PCB would enable a much more refined, compact design. The enclosure could be reduced in size, better shaped to the internal components, and more efficiently organized. Additionally, components such as the battery could be securely mounted rather than relying on friction-fit connections.

14.0 Development Challenges

14.1 Issues Encountered

Three main issues arose throughout the project's development. Table 14.1 describes the issues and their severity.

Table 14.1: Major issues encountered along with their severity and impact

Issue	Severity	Impact
Excessive Buzzer Current Draw	Medium	For battery powered operation, the voltage regulator is unable to provide ample current to the run system when the buzzer is drawing up to 50mA.
Consistency of positioning the receiver and emitter	High	Without proper positioning and blocking of external IR wavelengths, the sensor is extremely susceptible to noise from either sensor and finger movement, as well as other light sources, Making heartrate signals unreliable.
Determining optimal RC setup for average DC voltage conversion	Medium	In order to achieve accurate High and Low alarms a stable DC average voltage is required, with a higher RC, the output is more stable, but requires a long duration to settle, while a smaller RC gives fast settling with a less accurate DC conversion.

14.2 Troubleshooting Process

During troubleshooting, we tested the circuit one stage at a time rather than trying to solve everything at once. Using our test nodes, each section could be checked independently with the oscilloscope, making it easier to see where the signal was changing and which part of the circuit needed to be modified to improve our output.

For the buzzer problem, the circuit was tested under battery power, and it was found that the regulator couldn't properly handle the additional current draw while keeping the rest of the system stable. Based on our findings, we decided to use LED indicators rather than a buzzer, as they provided a more reliable output.

To address the sensor positioning issue, different finger placement orientations and light-shielding arrangements were tested to reduce noise from movement and external light. This led to changes to our enclosure setup, with the emitter and receiver fixed to provide a more reliable output.

For the average voltage conversion stage, different RC values were tested to find a balance between smooth output voltage and acceptable settling time. We tried multiple capacitor values and concluded that a 33 μF capacitor offers a good balance between settling time and a smooth output voltage.

14.3 Lessons Learned

Several important lessons were learned throughout the development of the project, particularly related to organization, communication and systematic testing.

One major lesson was the importance of project organization and team communication. As the circuit was being developed, many revisions were created, keeping updated files for each revision was crucial for working debugging during implementation.

Another important lesson was to test and implement stages of the circuit one at a time. Introducing multiple changes simultaneously made it difficult to identify the source of issues. Verifying each stage individually improved debugging and ensured proper operation before full system integration.

Finally, the team learned the importance component validation before insertion into the circuit. While testing various circuit stages, time was wasted debugging issues and trying to fix problems that stemmed from faulty components rather than the circuit design itself.

15.0 Experimental Results and Validation

15.1 Output Waveforms

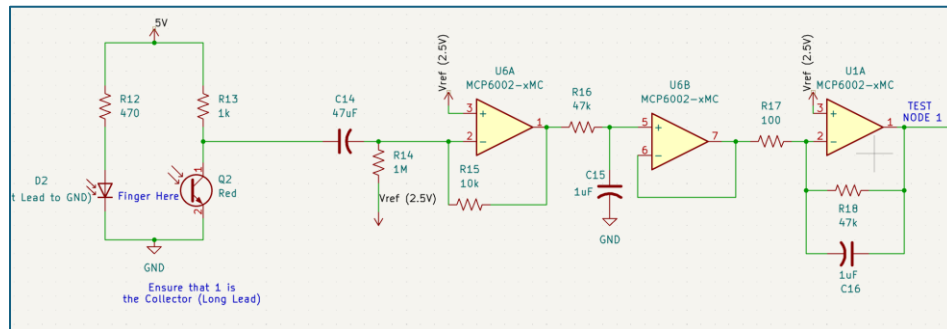


Figure 15.1: Circuit stage 1

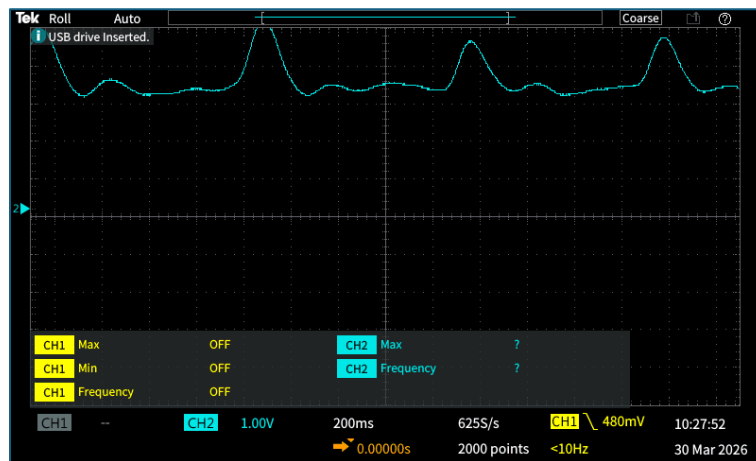


Figure 15.2: Output Waveform at Test Point 1

The output waveform captured in Figure 15.2 represents the signal where each periodic peak corresponds to a heartbeat. The large peaks indicate the systolic phase, caused by increased blood volume in the finger, while the smaller secondary peak is due to reflected pressure waves in the vascular system [4].

The signal has been conditioned using high-pass filtering to remove DC offsets from different users and low-pass filtering to reduce high-frequency noise. Amplification stages increase the small signal that we want to capture to a more usable level.

The resulting waveform is a cleaner, periodic analog signal centered around the 2.5 V reference, with peak spacing corresponding to the user's heart rate.

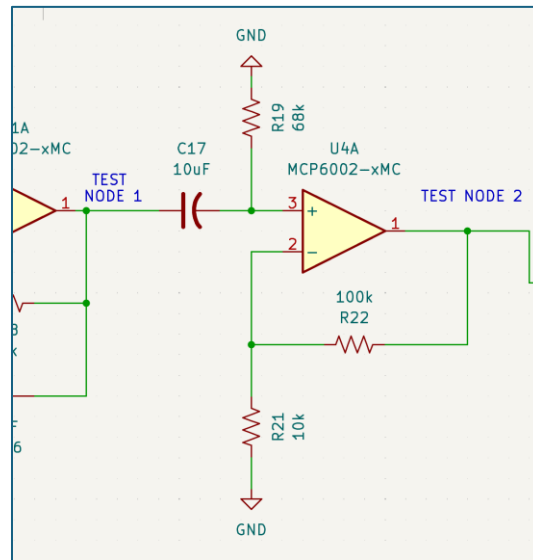


Figure 15.3: Stage 2 of Circuit

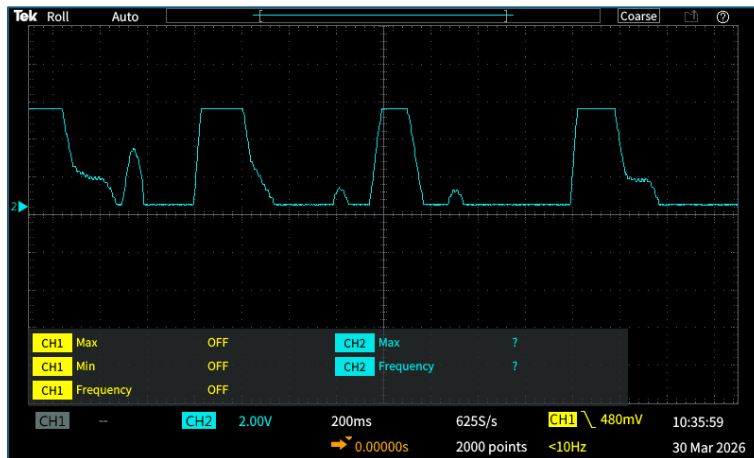


Figure 15.4: Output Waveform at Test Point 2

The output waveform captured in Figure 15.4 shows the AC-coupled op-amp stage, which removes the 2.5 V DC reference to center the signal. This stage also amplifies the signal to the

point where some pulses reach the op-amp rails, indicating saturation. The secondary peaks are also amplified but remain smaller in amplitude compared to the primary systolic peaks.

Additionally, the waveform is being clipped; however, this is acceptable, as the heart rate monitor only requires accurate pulse timing rather than amplitude information.

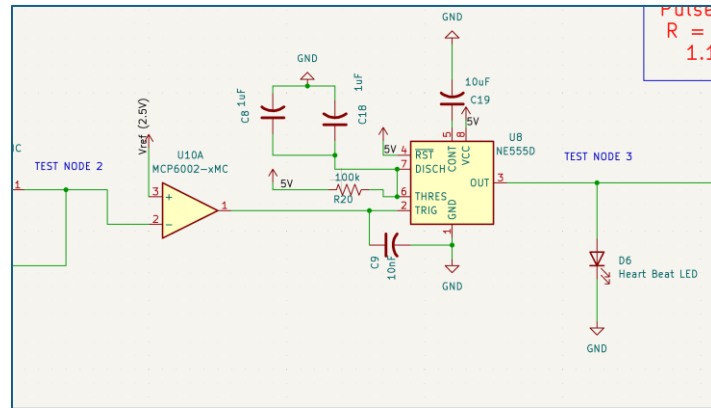


Figure 15.5: Stage 3 of Circuit

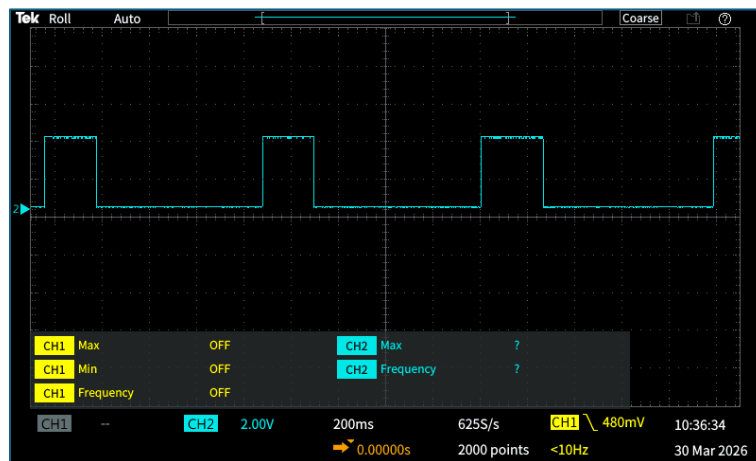


Figure 15.6: Output Waveform at Test Point 3

The next stage feeds the signal into a comparator referenced at 2.5 V as shown in Figure 15.5. The comparator converts the analog waveform into a digital square wave by switching its output whenever the input signal crosses the reference voltage. This effectively detects each heartbeat while rejecting small fluctuations and noise.

The resulting pulses are then passed into a 555-timer configured in monostable mode. The 555 timer standardizes the output by generating pulses of fixed width, determined by the RC time constant ($\tau = 1.1RC$). This ensures a consistent pulse as duration regardless of variations in the input signal shape or amplitude, producing clean and uniform square waves as shown in Figure 15.6.

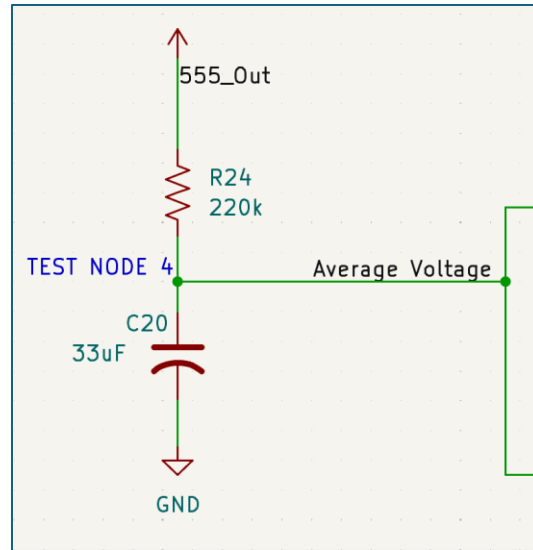


Figure 15.7: Stage 4 of Circuit

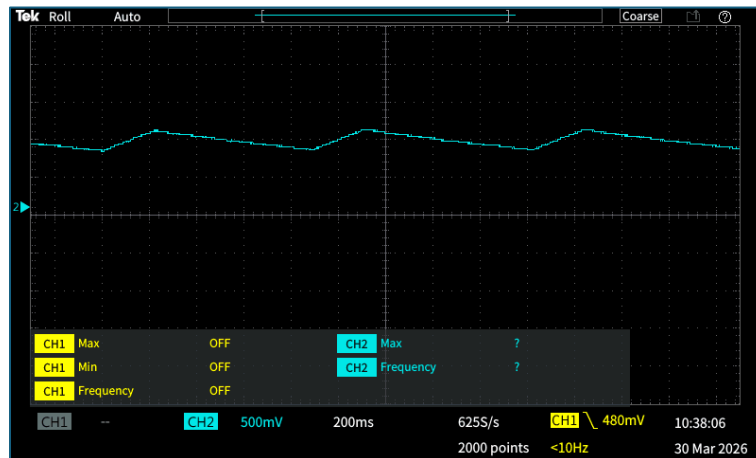


Figure 15.8: Output Waveform at Test Point 4

This output represents the averaged voltage used for high- and low-heart-rate detection. The signal is fed into two comparators, which determine whether the voltage exceeds or falls below the thresholds. A capacitor charges between heartbeats and discharges during each pulse, producing a voltage proportional to the time between beats and enabling heart rate classification

15.2 Response to Physical Activity

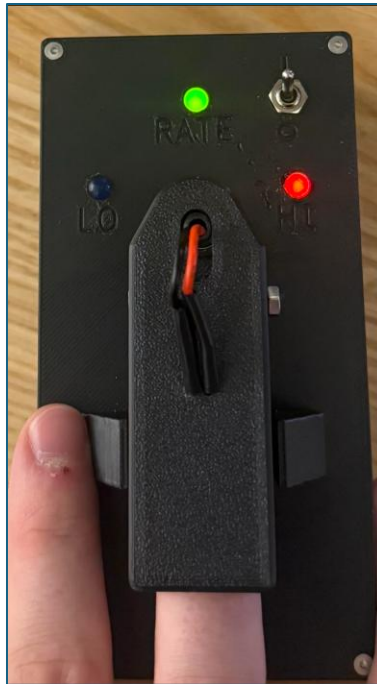


Figure 15.9: System output after physical activity

After the user performed jumping jacks and placed their finger into the slot, the system detected an elevated heart rate. When the user's heart rate exceeds the preset threshold, the red LED labeled "HI" illuminates. This threshold is set to approximately 125 BPM, indicating that the measured heart rate is above 125 BPM. This confirms that the system can accurately detect increases in heart rate and provide clear visual feedback to the user.

16.0 Teamwork and Contributions

16.1 Task Distribution and Individual Contributions

The project was completed through a collaborative team effort, with all members contributing to each stage of the design, implementation, and testing process. To improve efficiency and organization, major responsibilities were initially divided among team members based on strengths and project needs. These assignments reflect the primary contributors for each task; however, all team members remained actively involved throughout the development process and supported one another as needed.

The initial task distribution was organized as follows:

- I. Circuit Design: Ayden, James
- II. Circuit Implementation: Darion
- III. Enclosure Design and Implementation: Sam

Although these roles identified the primary contributors, the team worked closely together during troubleshooting, testing, and refinement stages. Design decisions were discussed collectively, and all members assisted in debugging, validation, and final system integration to ensure a cohesive and fully functional solution.

17.0 Conclusion

This project successfully demonstrated the design and implementation of an analog optical heart rate monitor using signal conditioning, filtering, and threshold detection techniques. Beginning with a small and noisy phototransistor signal, multiple amplification and filtering stages were used to extract a clean heartbeat waveform. The processed signal was converted into a square wave using a comparator and a 555 timer, allowing reliable frequency extraction and heart rate monitoring.

Threshold detection circuits were implemented to identify abnormal heart rate conditions, providing visual alerts. Testing under varying conditions showed that the system operated reliably, while calibration and long-term testing confirmed stability and repeatability.

Overall, the project demonstrated the effectiveness of modular analog signal processing for biomedical measurements. The final system met the project objectives and provided a functional, reliable heart rate monitoring solution, while also highlighting areas for future improvement such as enhanced noise reduction, improved enclosure design, and increased measurement accuracy.

18.0 References

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